



Microprocessor & Computer Architecture (μ pCA)

UE24CS251B

**Session
- 10**

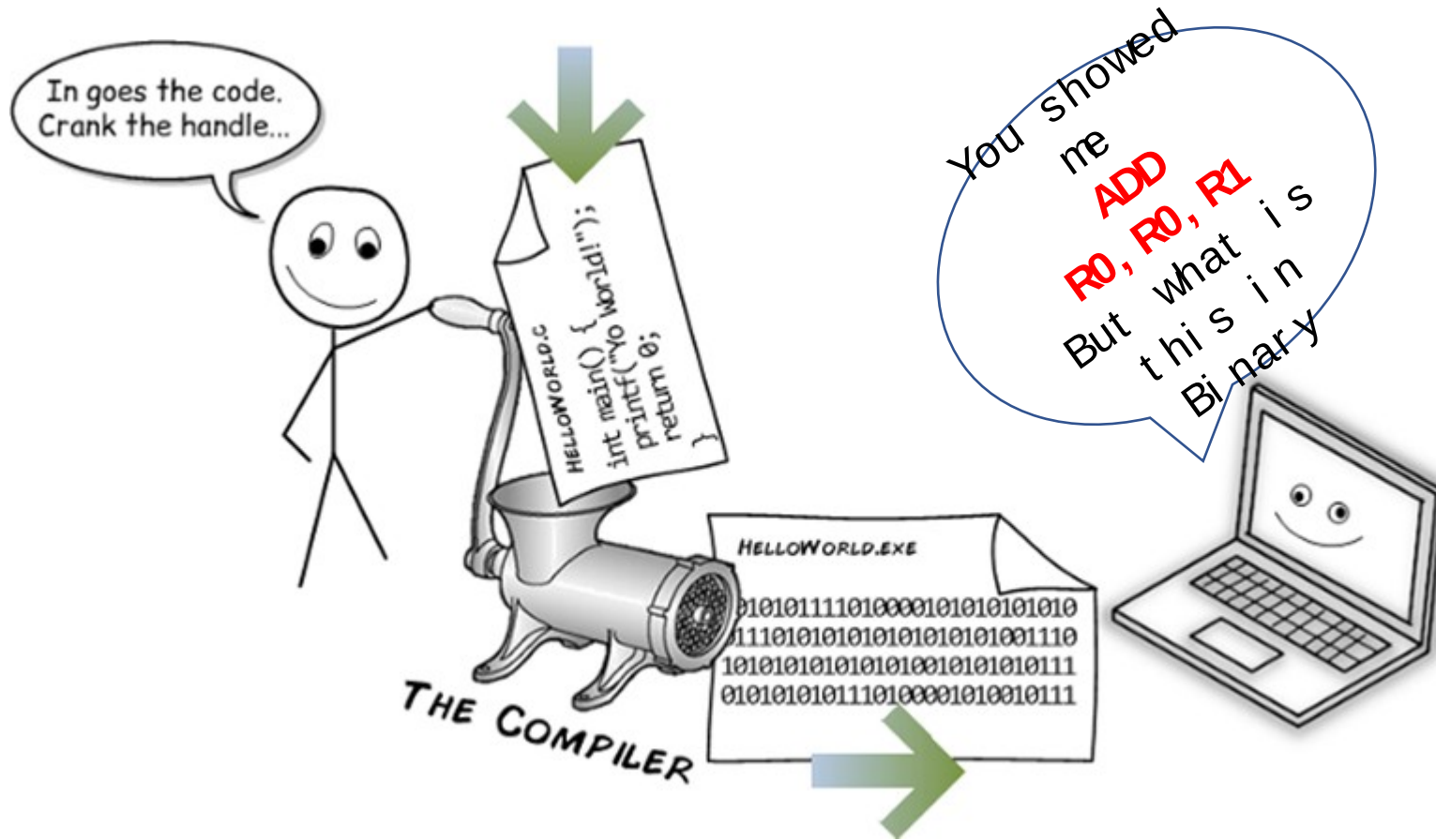


Microprocessor & Computer Architecture (μpCA)

UE24CS251B

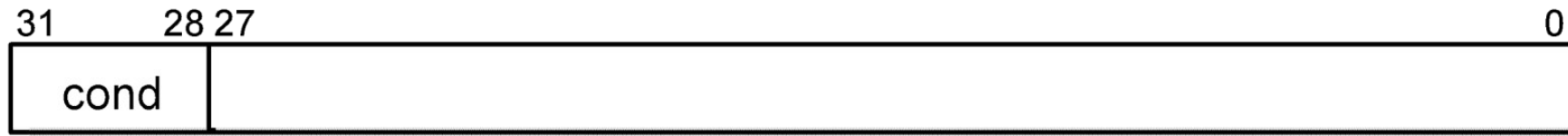
**Instruction
Encoding - 1**

Instruction Encoding

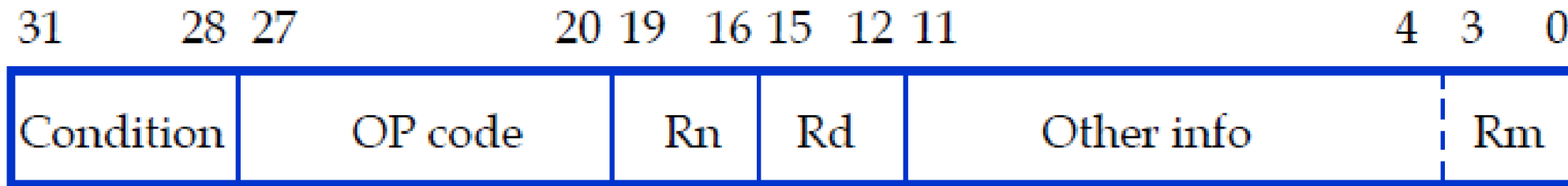


What is an ARM Instruction?

OPcode{ condition }{ S } Rd, Operand1, Operand2



Example: Data Processing Instruction

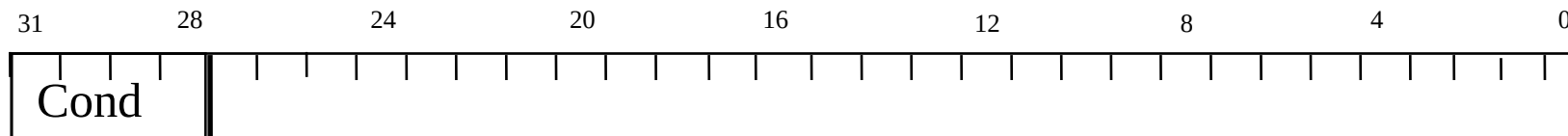


An instruction specifies a **Conditional Execution Code** (Condition), the **OP Code**, **Two or Three Registers** (Rn, Rd, and Rm), and some other information

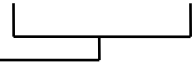
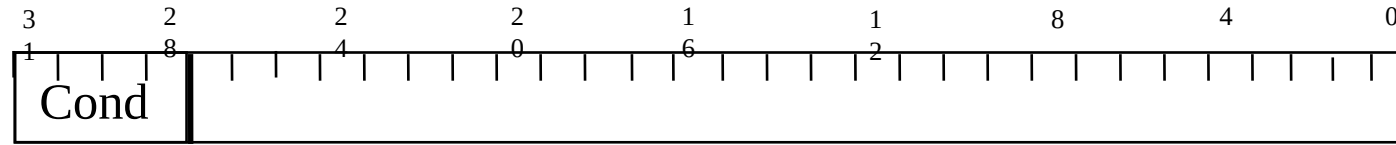
Condition

- Fifteen different conditions may be used: these are listed in **Condition code summary**.
- The **sixteenth (1111) is reserved**, and must not be used.
- In the **absence** of a suffix, the condition field of most instructions is set to "**Always**" (suffix **AL**). This means the instruction will always be executed regardless of the CPSR condition codes.

The ARM condition code field



The Condition Field



0000 = EQ - Z set (equal)

0001 = NE - Z clear (not equal)

0010 = HS / CS - C set (unsigned higher or same)

0011 = LO / CC - C clear (unsigned lower)

0100 = M - N set (negative)

0101 = PL - N clear (positive or zero)

0110 = VS - V set (overflow)

0111 = VC - V clear (no overflow)

1000 = HI - C set and Z clear (unsigned higher)

1001 = LS - C clear or Z (set unsigned lower or same)

1010 = GE - N set and V set, or N clear and V clear (> or =)

1011 = LT - N set and V clear, or N clear and V set (>)

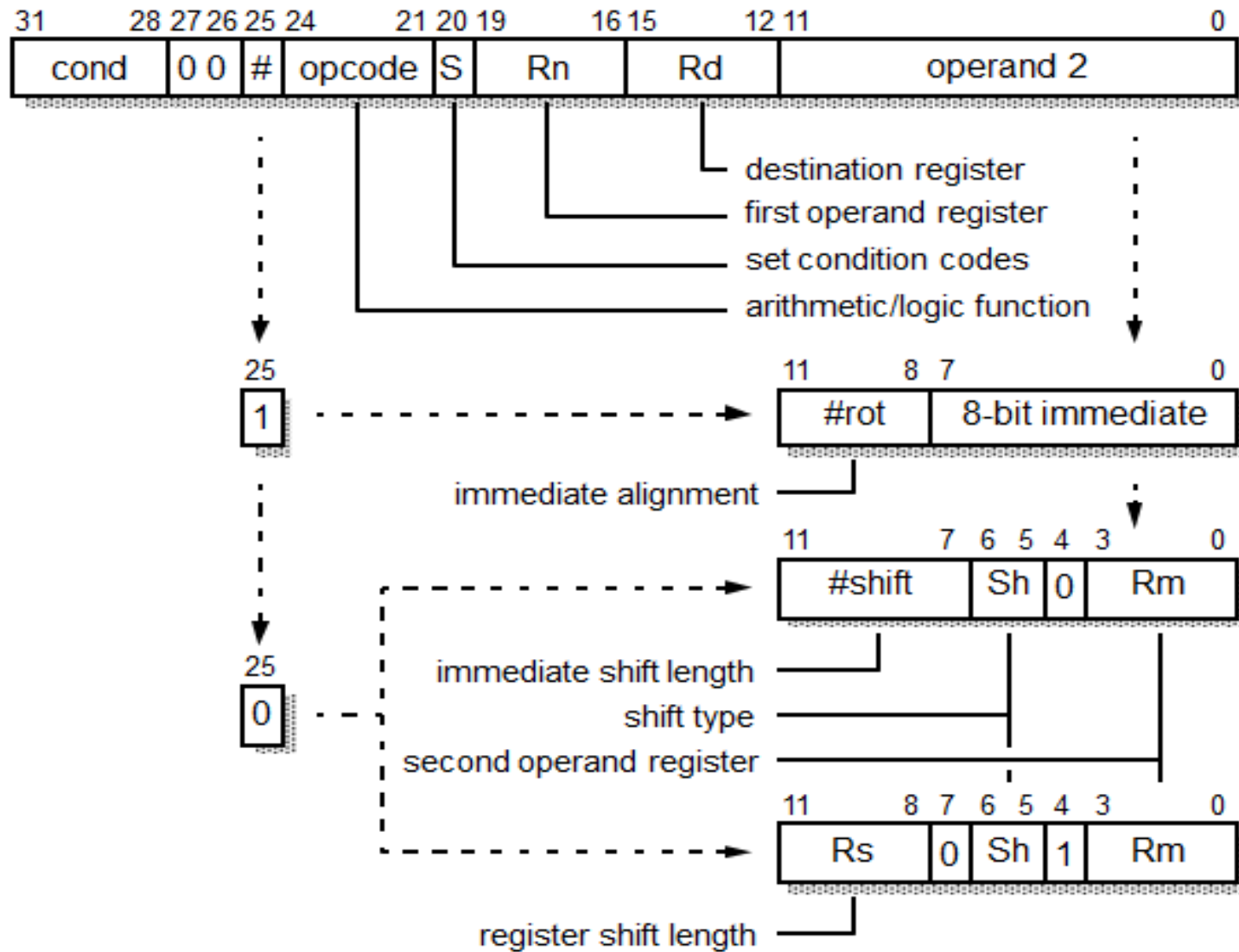
1100 = GT - Z clear, and either N set and V set, or N clear and V set (>)

1101 = LE - Z set, or N set and V clear, or N clear and V set (<, or =)

1110 = AL - always

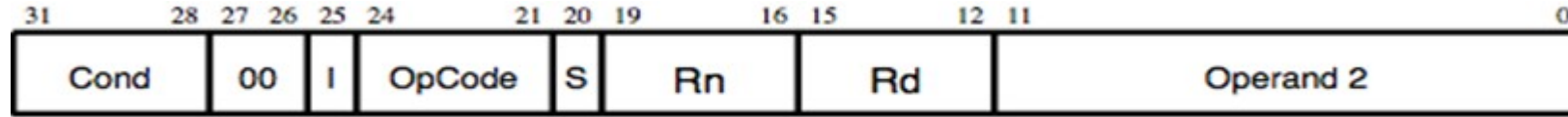
1111 = NV - reserved.

Data Processing Instruction



Opcode [24:21]	Mnemonic
0000	AND
0001	EOR
0010	SUB
0011	RSB
0100	ADD
0101	ADC
0110	SBC
0111	RSC
1000	TST
1001	TEQ
1010	CMP
1011	CMN
1100	ORR
1101	MOV
1110	BIC
1111	MVN

Data Processing Instruction - Example 01:



■ Ex:

ADD R1, R0, R2

1110 000 0100 S Rn Rd 00000000

Rm

1110 0000 1000 0000 0001 0000 0000 0010

Always

ADD

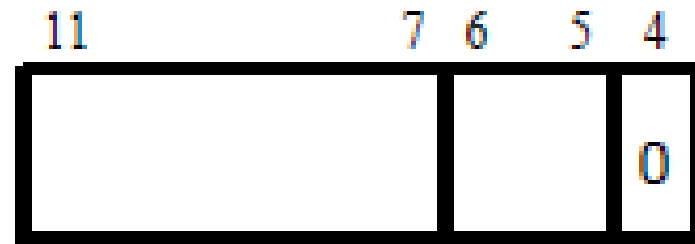
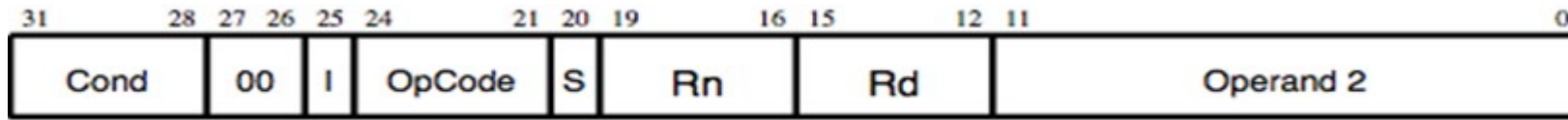
R0

R1

R2

E0801002

Data Processing Instruction



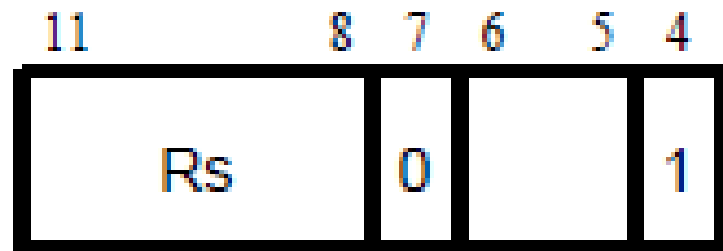
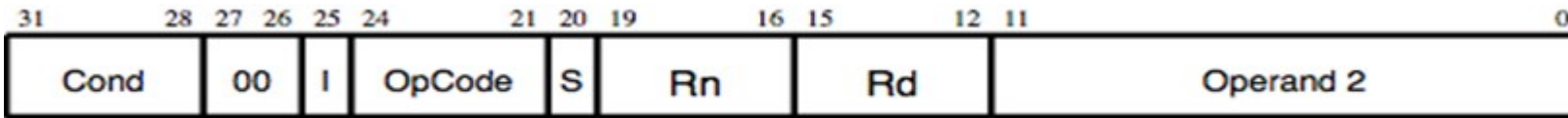
Shift type

- 00 = logical left
- 01 = logical right
- 10 = arithmetic right
- 11 = rotate right

Shift amount

5 bit unsigned integer

Data Processing Instruction



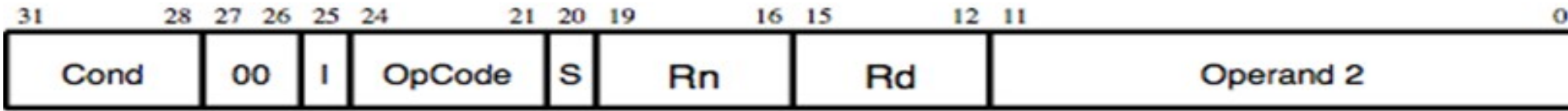
Shift type

- 00 = logical left
- 01 = logical right
- 10 = arithmetic right
- 11 = rotate right

Shift register

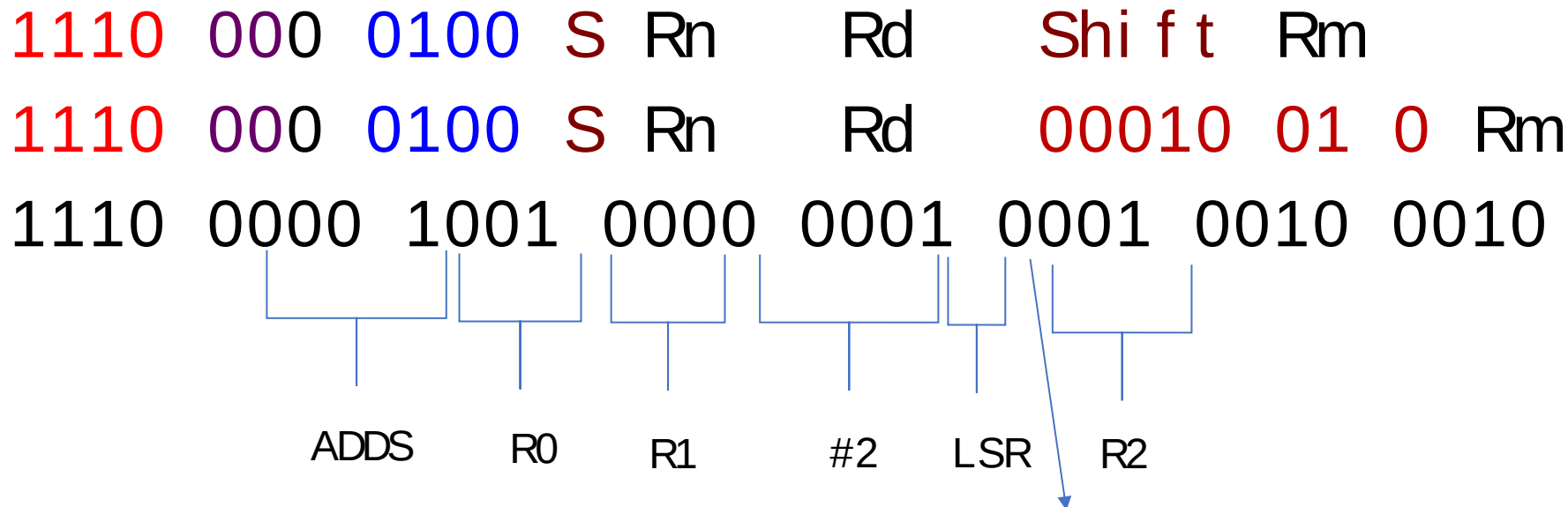
Shift amount specified in bottom byte of Rs

Data Processing Instruction - Example 02:



■Ex:

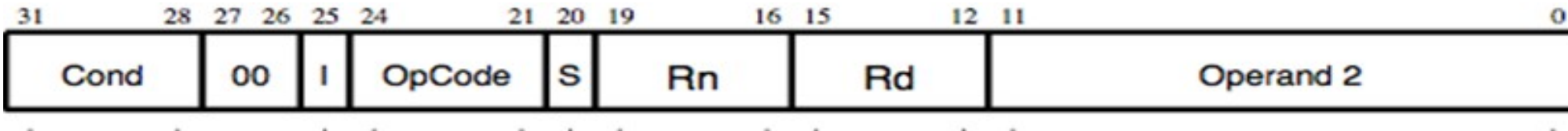
ADDS R1, R0, R2 LSR #2



E0901122

Shift is specified as
immediates

Data Processing Instruction Example 03:



■Ex:

ADDS R1, R0, R2 LSR R4

1110 000 0100 S Rn Rd Shift Rm
 1110 000 0100 S Rn Rd 0100 0 01 1 Rm
 1110 0000 1001 0000 0001 0100 0011 0010

└───┘ └───┘ └───┘ └───┘ └───┘ └───┘
 ADDS R0 R1 R4 LSR R2

Shift is specified in

Cond	0	0	1	Opcode				S	Rn		Rd		Operand 2											
Cond	0	0	0	0	0	0	A	S	Rd		Rn		Rs		1	0	0	1	Rm					
Cond	0	0	0	0	1	U	A	S	RdHi		RdLo		Rn		1	0	0	1	Rm					
Cond	0	0	0	1	0	B	0	0	Rn		Rd		0	0	0	0	1	0	0	1	Rm			
Cond	0	0	0	1	0	0	1	0	1	1	1	1	1	1	1	1	1	1	0	0	0	1	Rn	
Cond	0	0	0	P	U	0	W	L	Rn		Rd		0	0	0	0	1	S	H	1	Rm			
Cond	0	0	0	P	U	1	W	L	Rn		Rd		Offset			1	S	H	1	Offset				
Cond	0	1	1	P	U	B	W	L	Rn		Rd		Offset											
Cond	0	1	1														1							
Cond	1	0	0	P	U	S	W	L	Rn		Register List													
Cond	1	0	1	L	Offset																			

*Data Processing /
PSR Transfer*

Multiply

Multiply Long

Single Data Swap

Branch and Exchange

*Halfword Data Transfer:
register offset*

*Halfword Data Transfer:
immediate offset*

Single Data Transfer

Undefined

Block Data Transfer

Branch

A. 0000 B. 1111 C. 1110 D. 1000

A. Bits 24:21 B. Bits 19:16 C. Bits 15:12 D. Bits 11:0

A. 00 B. 01 C. 10 D. 11

1. If an instruction does not have a specific condition suffix (e.g., EQ, NE), it defaults to "Always" (AL). What is the 4-bit binary code for this "Always" condition?

- A. 0000 B. 1111 C. 1110 D. 1000

Answer – C. 1110

2. In the standard 32-bit ARM data processing instruction format, specific bits are allocated for the Destination Register (Rd). Which bit range corresponds to Rd?

- A. Bits 24:21 B. Bits 19:16 C. Bits 15:12 D. Bits 11:0

Answer – Bits 15:12

3. When using Immediate Shift in Operand 2, a 2-bit field specifies the type of shift to be performed. Which binary value represents a Logical Shift Right (LSR)?

- A. 00 B. 01 C. 10 D. 11

Answer – B. 01

4. In the instruction encoding format, bit 20 is labeled as 'S'. What is the function of this bit?
- A. It indicates that the instruction is a Store operation.
 - B. It specifies the size of the operand (byte or word).
 - C. It determines if the instruction updates the Condition Codes (CPSR).
 - D. It selects the Source register.
5. Based on the table of Opcodes (Bits 24:21) in the slides, which 4-bit binary sequence represents the MOV (Move) instruction?
- A. 0100
 - B. 1010
 - C. 1101
 - D. 1110

4. In the instruction encoding format, bit 20 is labeled as 'S'. What is the function of this bit?

- A. It indicates that the instruction is a Store operation.
- B. It specifies the size of the operand (byte or word).
- C. It determines if the instruction updates the Condition Codes (CPSR).
- D. It selects the Source register.

Answer - C. It determines if the instruction updates the Condition Codes (CPSR).

5. Based on the table of Opcodes (Bits 24:21) presented in the slides, which 4-bit binary sequence represents the MOV (Move) instruction?

- A. 0100
- B. 1010
- C. 1101
- D. 1110

Answer – C. 1101



THANK YOU



V R BADRI PRASAD

badri.prasad@pes.edu

Department of Computer Science and
Engineering